

Document Intelligence Platform

AI-powered extraction • financial validation • structured APIs • deployable dashboard

PDF / JPG / PNG

OCR + LLM

VALIDATION

REST API

Technical Case Study • Solution Presentation

Four financial document categories with controlled, machine-readable outputs.

From unstructured financial files to trusted data

The platform addresses the assessment's core challenge: complete extraction, validation, persistence and usable access.

THE CHALLENGE

- Different layouts and image quality.
- Scanned PDFs require OCR.
- Manual extraction is slow and inconsistent.
- Unreadable or missing values must not be invented.

THE SOLUTION

- Validate before expensive processing.
- OCR/text extraction + AI interpretation.
- Normalize into consistent structured JSON.
- Apply deterministic financial reconciliation.

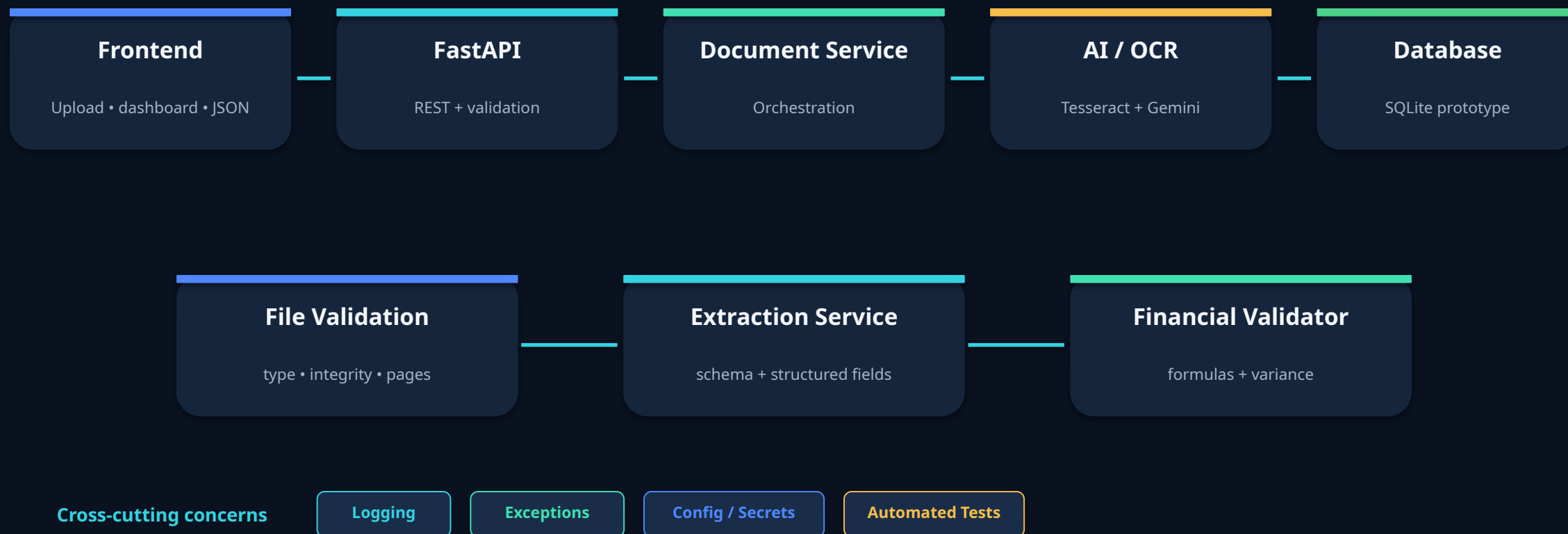
EVALUATOR EXPERIENCE

- Public frontend + REST API.
- Stored processing results in dashboard.
- Swagger/OpenAPI documentation.
- Readable results + raw JSON + status.

Reliable AI = extraction + grounding + deterministic validation + graceful failure.

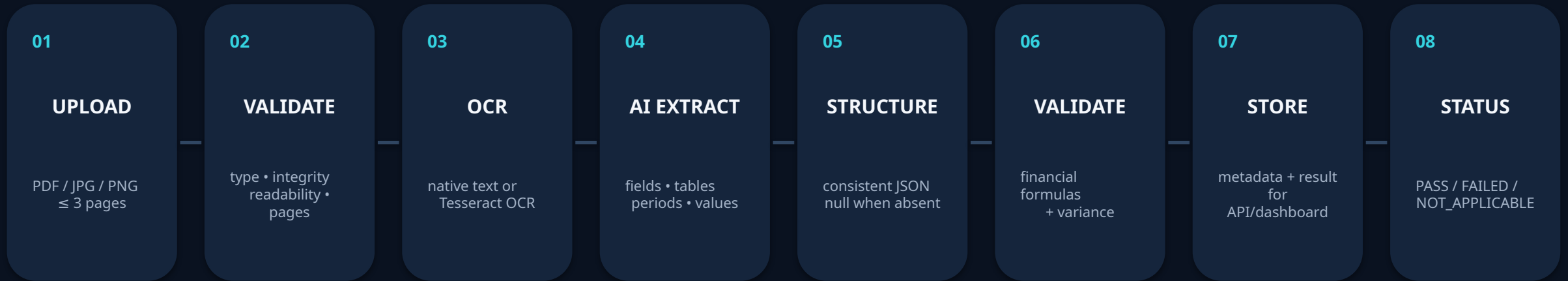
Modular pipeline with clear separation of concerns

Focused services keep file validation, OCR, AI extraction, financial rules, persistence and API behavior independently maintainable.



End-to-end document processing flow

The same controlled pipeline is used across Invoice, Balance Sheet, Profit & Loss and Cash Flow Statement inputs.



CONTROLLED FAILURE PATH

Invalid file, OCR failure, model failure, timeout or DB error → clear error response + logging; no stack traces or secrets exposed.

Grounded extraction instead of hard-coded answers

AI interprets OCR text; application logic then enforces a consistent schema and deterministic business rules.



EXTRACTION CONTRACT

- All meaningful visible fields.
- Financial line items + invoice tables.
- Comparative periods where present.
- Missing / unreadable → null.

GROUNDING

- Source text supports important values.
- Page number preferred when available.
- No unsupported inference.
- Confidence is optional.

PROVIDER STRATEGY

- Gemini selected for AI extraction.
- Free-tier usage keeps prototype cost-free.
- Secrets loaded from environment variables.
- Graceful model/API failure handling.

AI extraction is followed by deterministic reconciliation

Every check exposes a formula, inputs, calculated value, reported value, variance and status.

INVOICE

Qty × Unit Price ≈ Line Total
Subtotal + Tax – Discount ≈ Total

PASS • FAIL • NOT_APPLICABLE

BALANCE SHEET

Capital & Liabilities ≈ Assets
Component totals reconcile

PASS • FAIL • NOT_APPLICABLE

PROFIT & LOSS

Income – Expenditure ≈ Profit
Comparative periods checked

PASS • FAIL • NOT_APPLICABLE

CASH FLOW

Operating + Investing + Financing
+ FX ≈ Net Increase

PASS • FAIL • NOT_APPLICABLE

Example structured check

"formula"

: "subtotal + tax_amount - discount"

"calculated_value"

: 13125.00

"reported_value"

: 13125.00

"variance"

: 0.00

"status"

: "PASS"

Tolerance-based numeric comparison

0.01 absolute tolerance
+ relative tolerance for scale differences

One backend, two evaluation surfaces

The same stored result powers REST APIs and the browser dashboard, keeping behavior consistent.

REST API

GET

/api/v1/health

health check

POST

/api/v1/documents/process

upload + process

GET

/api/v1/documents

dashboard/list

GET

/api/v1/documents/{name}

latest result

EVALUATION DASHBOARD

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Processed Documents

Document	Type	Status	Time
invoice_01.jpg	Invoice	PASS	stored
balance_sheet.pdf	Balance	PASS	stored
cash_flow.pdf	Cash Flow	FAILED	stored

[Result view](#) • [extracted fields](#) • [tables](#) • [validation](#) • [raw JSON](#)

Built for evaluation, debugging and safe failure

Modular services, predictable outputs, secure configuration, logging and automated checks support maintainability.

4

supported document
types

3

page maximum enforced

8

processing stages

4

required API operations

0

hard-coded API secrets

VALIDATION & SECURITY

- PDF/JPG/PNG allow-list.
- Readability + page-count checks.
- Environment-variable secrets.
- Controlled HTTP errors.

OBSERVABILITY

- Processing-stage logging.
- OCR/model failure visibility.
- Exception handling without stack traces.
- Processing metadata retained.

TESTING & EVIDENCE

- Automated validation tests.
- API flow coverage.
- Scanned/image scenario.
- Failure / missing-value scenarios.

Submission-ready public interfaces

The prototype is deployed on Render with public source, API documentation and a browser-accessible application.

LIVE DEPLOYMENT

FRONTEND / API

<https://document-intelligence-platform-4d2l.onrender.com>

SWAGGER

<https://document-intelligence-platform-4d2l.onrender.com/docs>

GITHUB

<https://github.com/keerthishyamantula-maker/document-intelligence-platform>

Runtime

Docker + Python 3.12 + Tesseract OCR + FastAPI

DELIVERY CHECKLIST

- ✓ Public source repository
- ✓ Live frontend / backend
- ✓ Health + Swagger endpoints
- ✓ Architecture diagram
- ✓ Sample JSON outputs
- ✓ README + deployment docs
- ✓ Environment-based secrets

What the prototype demonstrates

A practical AI engineering pattern: use AI for interpretation, then use deterministic software where correctness can be enforced.

STRENGTHS

- End-to-end processing architecture.
- OCR support for scanned/image inputs.
- Structured extraction across four types.
- Financial reconciliation layer.
- Public API + dashboard + persistence.

KNOWN LIMITATIONS

- Gemini free-tier quota can limit live AI calls.
- SQLite on free hosting is prototype-grade rather than ideal durable production storage.
- OCR quality varies with source image quality.

PRODUCTION EVOLUTION

- Managed PostgreSQL.
- Queue-based asynchronous processing.
- Stronger document/OCR models.
- Field-level evidence + monitoring.
- Authentication, rate limits + object storage.

Key takeaway

AI extracts meaning. Software validates truth. APIs make it usable.

Thank you